Muhammad Talha Irfan

Multan, Pakistan | +92 314 6691351 | mtalirfan@gmail.com
mtalirfan.tech | linkedin.com/in/mtalirfan | github.com/mtalirfan

EDUCATION

National University of Sciences and Technology (NUST) — Islamabad, Pakistan
Bachelor of Mechanical Engineering – CGPA: 3.11 — *Nov 2021 – Jun 2025*
Elective Courses: Intro to Mechatronics Design, Robotics and Automation, Data Analytics for Engineers.

EXPERIENCE

PARCO - Pak-Arab Refinery Limited — Multan, Pakistan
Trainee Engineer — *Jan 2026 – Present*

Gourmet Pakistan — Multan, Pakistan
Management Trainee Production — *Sep 2025 – Nov 2025*
– Observed the working of the state-of-the-art mechatronic Krones equipment for beverage production, comprising of rotary mechanical equipment and PLC/HMI systems.
– Overview of the production facilities, processes, utilities, syrup and raw material for Gourmet's SKUs.

PARCO - Pak-Arab Refinery Limited — Multan, Pakistan
Mechanical Maintenance Intern — *Jun 2024 – Jul 2024*
– 2-day HSE training on Permit to Work, Risk Assessment, Life Saving Rules, Fire Safety and more.
– Spent 2 weeks each at Process, Workshop and U&OM Groups of Mechanical Maintenance.

PROJECTS

Dynamic Braille Generator System (Final Year Project) – Python, C++, Arduino, EasyEDA — *Sep 2024 – May 2025*
– Developed a Python program to convert text to grade 1 uncontracted and grade 2 contracted braille.
– Showcased a 5-cell LED display controlled by Arduino Mega, displaying input text into grade 1 braille.
– Designed a PCB Shield for Arduino Mega to evaluate the working of electro-mechanical braille cells.

Robot Manipulator – Robotics Toolbox, MATLAB, Python, NumPy, Matplotlib — *Oct 2024 – Nov 2024*
– Constructed the PUMA560 manipulator using the DH parameters in Robotics Toolbox.
– Plotted 3 unique Inverse Kinematics solutions for a set of random joint angles.
– Obtained the workspace of PUMA560, graphing 100000 end-effector positions.

Home Automation System – ESP8266, Proteus, C++, HTML, CSS, JavaScript — *Apr 2023 – May 2023*
– Implemented a NodeMCU-based DHT11 temperature sensor setup to control a DC fan and a relay.
– Developed a single-page web application, utilising HTTP requests and JavaScript DOM manipulation for manual control, and automation for a set temperature threshold.

CNC XY Plotter – Arduino, Proteus — *Nov 2022 – Dec 2022*
– Built an Arduino-based 2D CNC Plotter, with stepper motors controlling XY motion and a servo motor guiding a pen in Z axis, drawing various objects with precision on a horizontal surface.

SKILLS

– Mechanical and Electrical Design: AutoCAD | SOLIDWORKS | COMSOL Multiphysics | Proteus | LabVIEW | EasyEDA
– Embedded and Control Systems: C / C++ | Arduino | NodeMCU – ESP8266 | MATLAB – Simulink
– Robotics and Industrial Automation: Python | Webots | PLC Ladder Logic

– Data Science: Jupyter Notebook | Pandas | Scikit-Learn | Matplotlib | Seaborn | NumPy | Tensorflow – Keras | PyTorch
– Data Analytics and Automation: MS Excel | SQL | Jotform | Zapier | Make

– Miscellaneous: Technical Drawing | Technical Writing | Microsoft Office | 5S | Kaizen | Version Control – Git
– Soft Skills: Teamwork | Attention to Detail | Problem-Solving | Creativity
– Languages: English - Fluent | Urdu - Native

Courses and Certifications

– Data Analytics Essentials	Cisco Netacad	*Mar 2025*
– AI for Mechanical Engineers	Coursera	*Dec 2024*
– McKinsey Forward Program	McKinsey	*Dec 2024*
– Elements of AI: Introduction to AI and Building AI	MinnaLearn	*Jun 2024*
– Ethics and Communication for Engineers	ASME	*Nov 2023*
– Mastering 5S and Kaizen, The Toyota Way	UNAP	*Oct 2023*
– Introduction to Environmental Science Specialization	Coursera	*Jul 2023*
– Renewable Energy Specialization	Coursera	*Feb 2023*

VOLUNTEERING

NUST Community Services Club (NCSC) — Islamabad, Pakistan
Executive Publications — *Dec 2021 – Sep 2022*
– Transcribed positive changes on various social matters to captions for NCSC social media posts.